

Chapter I

Introduction

Fractal geometry will make you see everything differently. There is danger in reading further. You risk the loss of your childhood vision of clouds, forests, galaxies, leaves, feathers, flowers, rocks, mountains, torrents of water, carpets, bricks, and much else besides. Never again will your interpretation of these things be quite the same.

The observation by Mandelbrot [Mandelbrot 1982] of the existence of a “Geometry of Nature” has led us to think in a new scientific way about the edges of clouds, the profiles of the tops of forests on the horizon, and the intricate moving arrangement of the feathers on the wings of a bird as it flies. Geometry is concerned with making our spatial intuitions objective. Classical geometry provides a first approximation to the structure of physical objects; it is the language that we use to communicate the designs of technological products and, very approximately, the forms of natural creations. Fractal geometry is an extension of classical geometry. It can be used to make precise models of physical structures from ferns to galaxies. Fractal geometry is a new language. Once you can speak it, you can describe the shape of a cloud as precisely as an architect can describe a house.

This book is based on a course called “Fractal Geometry,” which has been taught in the School of Mathematics at the Georgia Institute of Technology for two years. The course is open to all students who have completed two years of calculus. It attracts both undergraduate and graduate students from many disciplines, including mathematics, biology, chemistry, physics, psychology, mechanical engineering, electrical engineering, aerospace engineering, computer science, and geophysical science. The delight of the students with the course is reflected in the fact there is now a second course, entitled “Fractal Measure Theory.” The courses provide a compelling vehicle for teaching beautiful mathematics to a wide range of students.

Here is how the course in Fractal Geometry is taught. The core is Chapter II, Chapter III, sections 1–5 of Chapter IV, and sections 1–3 of Chapter V. This is followed by a collection of delightful special topics, chosen from Chapters VI, VII, and VIII. The course is taught in 30 one-hour lectures.

Chapter II introduces the basic topological ideas that are needed to describe subsets to spaces such as $\mathbb{R}^2$. The framework is that of metric spaces; this is adopted because metric spaces are both rigorously and intuitively accessible, yet full of surprises. They provide a suitable setting for fractal geometry. The concepts introduced include openness, closedness, compactness, convergence, completeness, connectedness, and equivalence of metric spaces. An important theme concerns properties that are preserved under equivalent metrics. Chapter II concludes by presenting the most exciting idea: a metric space, denoted $\mathcal{H}$, whose elements are the nonempty compact subsets of a metric space. Under the right conditions this space is complete, sequences converge, and fractals can be found!

Chapter III deals with transformations on metric spaces. First, the goal is to develop intuition and practical experience with the actions of elementary transformations on subsets of spaces. Particular attention is devoted to affine transformations and Möbius transformations in $\mathbb{R}^2$. Then the contraction mapping principle is revealed, followed by the construction of contraction mappings on $\mathcal{H}$. Fractals are discovered as the fixed points of certain set maps. We learn how fractals are generated by the application of “simple” transformations on “simple” spaces, and yet they are geometrically complicated. We explain what an iterated function system (IFS) is, and how it can define a fractal. Iterated function systems provide a convenient framework for the description, classification, and communication of fractals. Two algorithms, the “Chaos Game” and the Deterministic Algorithm, for computing pictures of fractals are presented. Attention is then turned to the inverse problem: given a compact subset of $\mathbb{R}^2$, fractal, how do you go about finding a fractal approximation to it? Part of the answer is provided by the Collage Theorem. Finally, the thought of the wind blowing through a fractal tree leads to discovery of conditions under which fractals depend continuously on the parameters that define them.

Chapter IV is devoted to dynamics on fractals. The idea of addresses of points on certain fractals is developed. In particular, the reader learns about the metric space to which addresses belong. Nearby addresses correspond to nearby points on the fractal. This observation is made precise by the construction of a continuous transformation from the space of addresses to the fractal. Then dynamical systems on metric spaces are introduced. The ideas of orbits, repulsive cycles, and equivalent dynamical systems are described. The concept of the shift dynamical system associated with an IFS is introduced and explored. This is a visual and simple idea in which the author and the reader are led to wonder about the complexity and beauty of the available orbits. The equivalence of this dynamical system with a corresponding system on the space of addresses is established. This equivalence takes no account of the geometrical complexity of the dance of the orbit on the fractal. The chapter then moves towards its conclusion, the definition of a chaotic dynamical system and the realization that “most” orbits of the shift dynamical system on a fractal are chaotic. To this end, two simple and delightful ideas are shown to the reader. The Shadow Theorem

illustrates how apparently random orbits may actually be the “shadows” of deterministic motions in higher-dimensional spaces. The *Shadowing Theorem* demonstrates how a rottenly inaccurate orbit may be trailed by a precise orbit, which clings like a secret agent. These ideas are used to make an explanation of why the “Chaos Game” computes fractals.

Chapter V introduces the concept of fractal dimension. The fractal dimension of a set is a number that tells how densely the set occupies the metric space in which it lies. It is invariant under various stretchings and squeezings of the underlying space. This makes the fractal dimension meaningful as an experimental observable; it possesses a certain robustness and is independent of the measurement units. Various theoretical properties of the fractal dimension, including some explicit formulas, are developed. Then the reader is shown how to calculate the fractal dimension of real-world data, and an application to a turbulent jet exhaust is described. Lastly the Hausdorff-Besicovitch dimension is introduced. This is another number that can be associated with a set. It is more robust and less practical than the fractal dimension. Some mathematicians love it; most experimentalists hate it; and we are intrigued.

Chapter VI is devoted to fractal interpolation. The aim of the chapter is to teach the student practical skill in using a new technology for making complicated curves and fitting experimental data. It is shown how geometrically complex graphs of continuous functions can be constructed to pass through specified data points. The functions are represented by succinct formulas. The main existence theorems and computational algorithms are provided. The functions are known as fractal interpolation functions. It is explained how they can be readily computed, stored, manipulated and communicated. “Hidden variable” fractal interpolation functions are introduced and illustrated; they are defined by the shadows of the graphs of three-dimensional fractal paths. These geometrical ideas are extended to introduce space-filling curves.

Chapter VII gives an introduction to Julia sets, which are deterministic fractals that arise from the iteration of analytic functions. The objective is to show the reader how to understand these fractals, using the ideas of Chapters III and IV. In so doing we have the pleasure of explaining and illustrating the Escape Time Algorithm. This algorithm is a means for computergraphical experimentation on dynamical systems that act on two-dimensional spaces. It provides illumination and coloration, a seachlight to probe dynamical systems for fractal structures and regions of chaos. The algorithm relies on the existence of “repelling sets” for continuous transformations which map open sets to open sets. The applications of Julia sets to biological modelling and to understanding Newton’s method are considered.

Chapter VIII is concerned with how to make maps of certain spaces, known as parameter spaces, where every point in the space corresponds to a fractal. The fractals depend “smoothly” on the location in the parameter space. How can one make a picture that provides useful information about what kinds of fractals are located where? If both the space in which the fractals lie and the parameter space

are two-dimensional, the parameter space can sometimes be “painted” to reveal an associated Mandelbrot set. Mandelbrot sets are defined, and three different examples are explored, including the one discovered by Mandelbrot. A computergraphical technique for producing images of these sets is described. Some basic theorems are proved.

Chapter IX is an introduction to **measures on fractals and to measures in general**. The chapter is an outline that can be used by a professor as the basis of a course in fractal measure theory. It can also be used in a standard measure theory course as a source of applications and examples. One goal is to demonstrate that measure theory is a workaday tool in science and engineering. Models for real-world images can be made using measures. The variations in color and brightness, and the complex textures in a color picture, can be successfully modelled by measures that can be written down explicitly in terms of succinct “formulas.” These measures are desirable for image engineering applications, and have a number of advantages over non-negative “density” functions. Section 1 provides an intuitive description of measures and motivates the rest of the chapter. The context is that of Borel measures on compact metric spaces. Fields, sigma-fields, and measures are defined. Carathéodory’s extension theorem is introduced and used to explain what a Borel measure is. Then the integral of a continuous real-valued function, with respect to a measure, is defined. The reader learns to evaluate some integrals. Next the space $\mathcal{P}$ of normalized Borel measures on a compact metric space is defined. With an appropriate metric, $\mathcal{P}$ becomes a compact metric space. Succinctly defined contraction mappings on this space lead to measures that live on fractals. Integrals with respect to these measures can be evaluated with the aid of Elton’s ergodic theorem. The book ends with a description of the application of these measures to computer graphics.

This book teaches the tools, methods, and theory of *deterministic* geometry. It is useful for describing *specific* objects and structures. Models are represented by succinct “formulas.” Once the formula is known the model can be reproduced. We do not consider statistical geometry. The latter aims at discovering general statistical laws that govern families of similar-looking structures, such as *all* cumulus clouds, *all* maple leaves, or *all* mountains.

In deterministic geometry, structures are defined, communicated, and analyzed, with the aid of elementary transformations such as affine transformations, scalings, rotations, and congruences. A fractal set generally contains infinitely many points whose organization is so complicated that it is not possible to describe the set by specifying directly where each point in it lies. Instead, the set may be defined by “the relations between the pieces.” It is rather like describing the solar system by quoting the law of gravitation and stating the initial conditions. Everything follows from that. It appears always to be better to describe in terms of relationships.